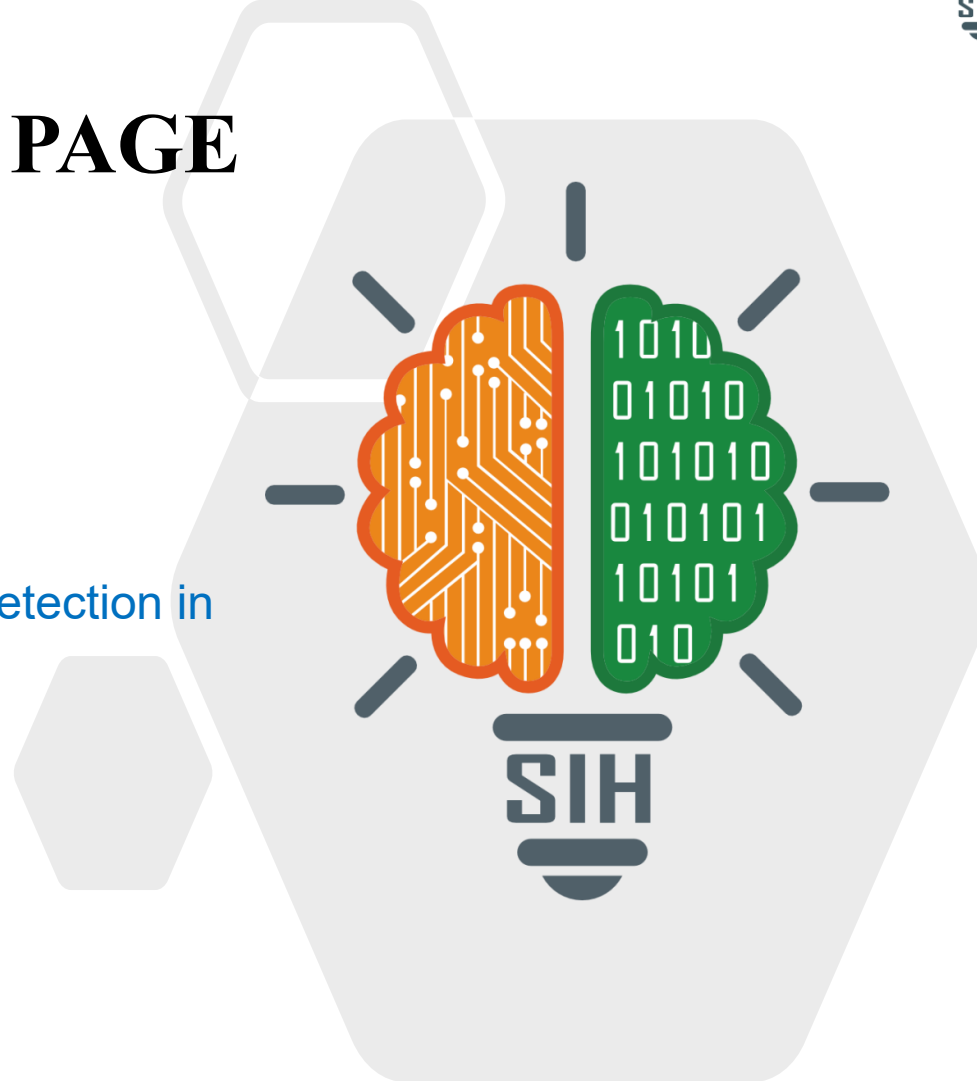


TITLE PAGE

AgniDrishti

Catch the drifting in-spec part at hour 24, not 168.

- **Problem Statement ID** – 26170
- **Problem Statement Title**- AI-Driven Anomaly Detection in Component Burn-In & Screening
- **Theme**- Smart Automation
- **PS Category**- Software
- **Team ID**- (to be assigned on the SIH portal)
- **Team Name**- Regnum Carya



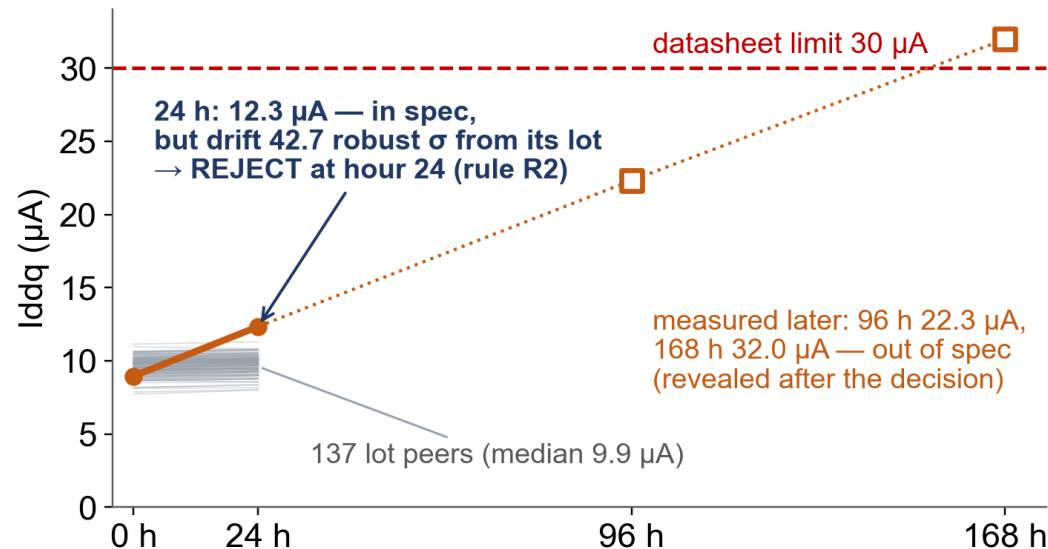
In spec is not the same as healthy

Proposed Solution (Describe your Idea/Solution/Prototype)

Detailed explanation of the proposed solution

A burn-in part can pass its datasheet limit yet be abnormal for its own lot or drifting toward failure — the latent defect the PS targets. At the 24 h checkpoint AgniDrishti compares every part with its lot peers, forecasts its 168 h value with a calibrated range, and returns **PASS / REVIEW / REJECT** with the rule that fired.

REAL RUN · SYNTHETIC held-out test lot · the datasheet check says PASS



Innovation and uniqueness of the solution

Drifting rows caught at 24 h · held-out test lots

4 / 30

Datasheet limit only

4 / 30

+ DPAT lot level

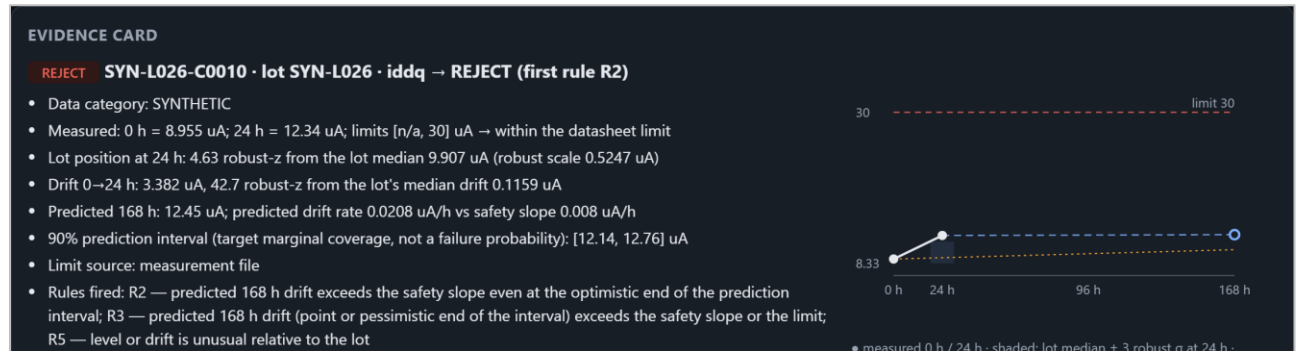
20 / 30

+ lot-relative drift

24 / 30AgniDrishti · 10.4 % to
REVIEW

Doubt is never a PASS — bad data, or a 168 h range that could reach the limit, goes to an engineer. Not a weighted risk score: the rule that fired is the explanation. Every layer stays only if it helps on unseen lots.

SAME PART IN THE PROTOTYPE · screenshot of the evidence card



How it addresses the problem

Module A level and 0→24 h drift vs the part's own lot (robust z)

Module B V0h + V24h → V168h forecast with a 90 % calibrated range

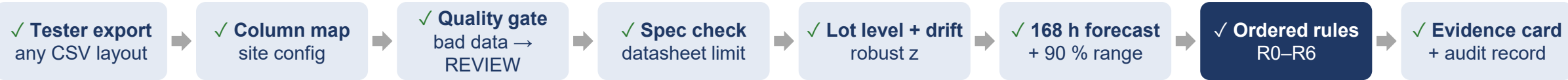
Decision ordered rules R0–R6 — doubt goes to REVIEW, never PASS

Every layer must beat a simpler baseline on unseen lots — or it is removed.

Technologies to be used (e.g. programming languages, frameworks, hardware)

Python 3.11 · pandas / NumPy · robust statistics · least-squares forecasts · split-conformal ranges · web service + UI · pytest · YAML config — offline, CPU only.
Learned from historical lots: forecast model chosen by held-out MAE, 90 % range calibrated on held-out lots, review threshold set on validation lots.

Methodology and process for implementation (Flow Charts/Images/ working prototype)



Decision rules — first match wins

R0	bad or missing data	REVIEW
R1	outside the datasheet limit now	REJECT
R2	drift rate > safety slope, even at the optimistic end of the range	REJECT
R3	point or pessimistic end reaches the slope or the limit	REVIEW
R4	level and drift both ≥ 6 robust σ from the lot (DPAT)	REJECT
R5	level or drift ≥ 2.5 robust σ from the lot	REVIEW
R6	none of the above	PASS

robust z = $(x - \text{lot median}) / (\text{IQR} / 1.35)$ **drift rate** $r = (\hat{Y}_{168} - V_0) / 168 \text{ h}$
safety slope $s = \Delta_{\text{allow}} / 168 \text{ h}$; $\Delta_{\text{allow}} = 15 \%$ of V_0 per parameter (site config; PS leaves it undefined)
PS scoring: REVIEW and REJECT both count as flagged — a missed defect costs more than a review.

Measured: held-out test lots, SYNTHETIC, evaluated once

Cumulative layers (E1–E6)	Recall	Review rate	False reject
E1 datasheet limit	0.133	1.2 %	0.0 %
E2 + DPAT lot level	0.133	1.9 %	0.0 %
E3 + lot-relative drift	0.667	3.5 %	0.3 %
E4 + 168 h forecast	0.667	3.3 %	0.3 %
E5 + 90 % range	0.667	3.4 %	0.3 %
E6 full ordered rules	0.800	10.4 %	0.3 %

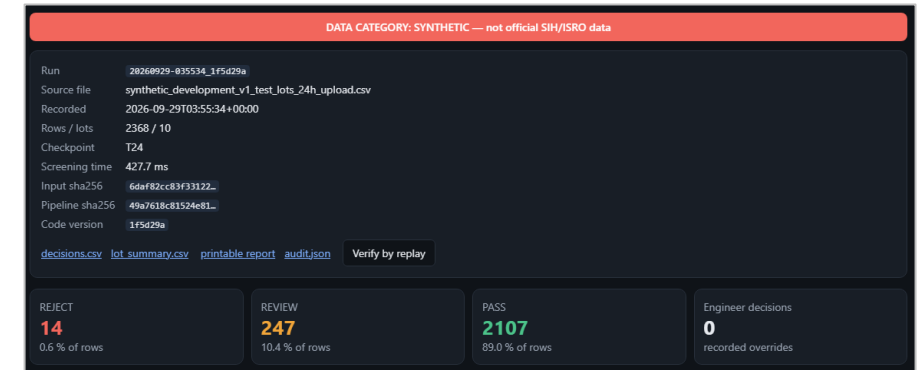
168 h forecast MAE beats persistence (validation lots): Iddq 0.297 vs 0.338 μA · tpd 0.047 vs 0.050 ns. Range coverage on test 0.887 (target 0.90).

Honest finding: the gain comes from lot drift; forecast and range added no detection here. Recall target 0.95 not yet met; 30 positives, so rates are imprecise.

No LSTM: 2–4 readings per part — sequence models add opacity, not accuracy.

Potential challenges and risks	Strategies for overcoming these challenges
No official dataset for PS 26170	Synthetic PS-shaped lots, always labelled; the site-fit script recalibrates on ISRO history (built)
Hidden ground truth; scoring metric undefined	REVIEW + REJECT count as flagged; report recall at a fixed review budget and 168 h MAE vs persistence
“You detect the defects you injected”	2 held-out behaviour families sealed by SHA-256 on 26 Sep 2026, before detector code; test lots evaluated once; leakage tests
Whole lot degrades / contaminated lot	Median / IQR statistics, plus absolute-limit and safety-slope backstops
Safety slope not defined in the PS	Δ allow per parameter in site config (default 15 % of V0); sensitivity grid reported
Late-onset defects look healthy at 24 h (6 of 30 test escapes)	Stated limit; re-screen when the 96 h reading arrives (next)
One glitchy 24 h reading → false REJECT (all 6 test false rejects)	Proposed v2: single-jump cases go to REVIEW, not REJECT — to be tested on fresh pre-registered data
Different tester formats and units	Column mapping + unit conversion from a site config file (built)

Analysis of the feasibility of the idea



Live service · run 20260929-035534 · replay → identical decisions

Technical Standard PC, CPU only: 2,368 rows screened in 0.43 s (one recorded run).

Operational Engineer works a ranked REVIEW queue; overrides are logged beside the pipeline decision.

Financial Open-source stack, no licence or cloud; data never leaves the facility.

Built and verified

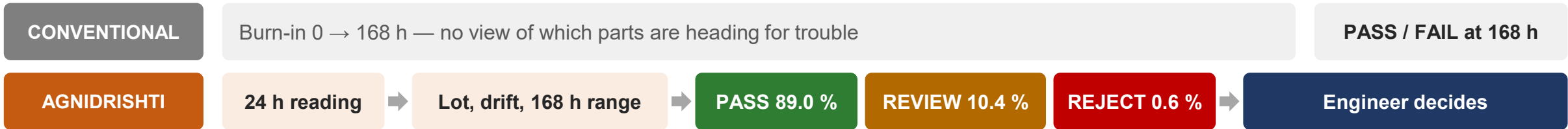
✓ all 7 layers ✓ web UI + CLI ✓ any-CSV ingest
✓ 111 automated tests pass ✓ audit record + replay

Code: github.com/Mrudula211/AgniDrishti

Decide at hour 24 — with evidence

Potential impact on the target audience

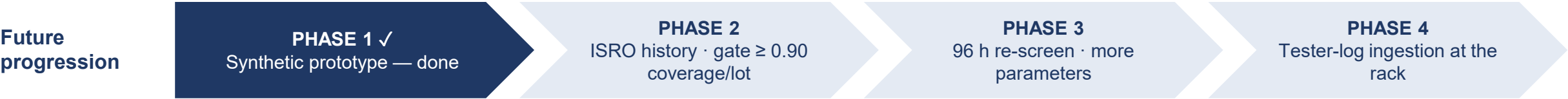
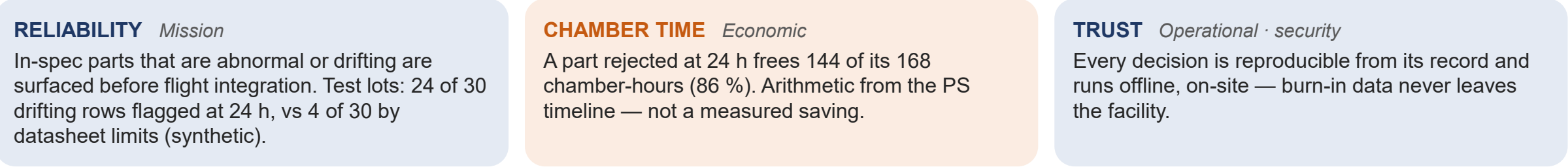
For: QA and reliability engineers screening space-grade electronics



Decision shares measured on the held-out SYNTHETIC test lots (2,368 rows).

At the 24 h checkpoint the QA engineer gets a ranked REVIEW queue — each part with its numbers, its lot, its forecast range and the rule that fired — and records a decision with a reason.

Benefits of the solution (social, economic, environmental, etc.)



Gains on ISRO hardware will be measured after validation on representative ISRO data.

Details / Links of the reference and research work

OFFICIAL

SIH 2026 PS 26170 — AI-Driven Anomaly Detection in Component Burn-In & Screening (ISRO). sih.gov.in/sih2026PS
SIH 2026 Guidelines; SIH 2026 Idea Presentation Format — sih.gov.in/letters/2026/

RELIABILITY PRACTICE (informed the design — no compliance claimed)

AEC-Q001 Rev-D, Guidelines for Part Average Testing, AEC, 2011 — our Module A baseline (DPAT)
ESCC Generic Specification No. 9000, Issue 10, ESA, 2018 — burn-in drift limits
ESCC Detail Specification No. 9202/045, Issue 6, ESA, 2020 — example per-parameter drift limits (IDD ± 75 nA; IOL ± 15 %)

DATA (external public dataset — not SIH/ISRO data)

NASA PCoE Data Repository — MOSFET Thermal Overstress Aging (Celaya, Saxena, Saha, Goebel)
NASA PCoE — IGBT Accelerated Aging (Celaya, Wysocki, Goebel, 2009)
Downloaded and hashed; planned methodology check only — not yet run.

PRIOR ART

Burn-in ML: I. Ahmed, P. Baraldi, E. Zio, H. Lewitschnig, Comput. Ind. Eng. 204, 111115, 2025 · doi:10.1016/j.cie.2025.111115
Lot health for burn-in: L. Langenberg, R. Pätzold, A. Khalid, ESREL 2023
Burn-in outliers: US 8,010,310 B2, Method and apparatus for identifying outliers following burn-in testing, 2011

METHOD

Conformal prediction: J. Lei et al., JASA 113(523), 2018 · doi:10.1080/01621459.2017.1307116
Grouped (lot) data: R. Dunn, L. Wasserman, A. Ramdas, JASA 118(544), 2023 · doi:10.1080/01621459.2022.2060112
Degradation modelling: C. J. Lu, W. O. Meeker, Technometrics 35(2), 1993 · doi:10.1080/00401706.1993.10485038
Reject option (REVIEW): C. K. Chow, IEEE Trans. Inf. Theory 16(1), 1970 · doi:10.1109/TIT.1970.1054406

OUR OWN EVIDENCE — CHECK IT

Code: github.com/Mrudula211/AgniDrishti · branch p1-synthetic-data @ 1f5d29a · 111 automated tests · E1–E6 pre-registered before detector code
Held-out behaviour families sealed 26 Sep 2026 (SHA-256 1a7de276...) before model tuning — the defence against “you detect what you injected”.